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Predicting transfer from a game-based learning environment

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ABSTRACT

This study examined the predictive impact of variables from self-regulated learning models on transfer following multiple gameplay sessions in a classroom game-based learning environment (GBLE). The game focused on science curriculum including map models, map navigation, and landforms. Fifth-grade students (N = 594) interacted with CRYSTAL ISLAND - UNCHARTED DISCOVERY during six 50-min sessions over four weeks integrated with classroom instruction. The transfer activity required students to create an island on grid paper that included at least seven different landforms, a map navigation activity created for peers, and an attached map scale model. Results of multiple regression analyses revealed that when other variables were accounted for, none of the measured motivational variables including interest for the game, performance-approach goal orientation, mastery-approach goal orientation, or self-efficacy for science were significant positive predictors of transfer. Interestingly, video game self-efficacy was a significant negative predictor of transfer performance. Prior knowledge and change in science knowledge after gameplay were also significant predictors of transfer. Girls performed as well as boys on the transfer task despite completing less game quests and had significantly lower posttest content scores. Interest was a consistent predictor of performance for girls but not boys for in-game performance, posttest content scores, and transfer scores. Regression models were systematically less able to predict performance moving from content knowledge, to in-game performance, to transfer. Implications for the design of GBLEs to facilitate transfer are discussed.

1. Introduction

Recent evidence suggests that digital game-based learning environments (GBLEs) can have a range of positive cognitive, affective, and motivational benefits (Clark, Tanner-Smith, & Killingsworth, 2016; Connolly, Boyle, MacArthur, Hainey, & Boyle, 2012; Tsai & Tsai, 2018). GBLEs are touted because of the engaging nature by which they approach instruction (Prensky, 2001). Currently, however, there is a major challenge for researchers of GBLEs to demonstrate transfer of content and strategies outside of the game environment (Hung & Van Eck, 2010; Mayer, 2011; Nietfeld, 2018). Yet, most studies of GBLEs are limited in their duration leaving little potential to achieve transfer (Jackson & McNamara, 2013). Moreover, evidence for transfer has come mostly from near-transfer test items or responses taken directly after gameplay with a high degree of alignment with the game itself (Pilegard & Mayer, 2016).

In order to demonstrate transfer outside of a game setting, students must effectively establish, monitor, and regulate academic goals that are aligned with curricular objectives. This process requires effective self-regulated learning (SRL), or self-generated thoughts, feelings, and actions for attaining one's learning goals (Zimmerman and Moylan, 2009), and encompasses the coordination of motivational, cognitive, and metacognitive processes. In traditional learning environments self-regulation has been linked with the

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transfer of learning (Bannert & Reimann, 2012; Fuchs et al., 2003; Glaser & Brunstein, 2007; Zepeda, Richey, Ronevich, & Nokes-Malach, 2015). However, in GBLEs the research in this regard is lacking. GBLEs have a high degree of potential to encourage both SRL and transfer, yet they are particularly susceptible to extraneous processing that detracts from the desired learning experience (Mayer, 2019). A pillar of SRL is the autonomy the learner is afforded and the opportunity to make critical choices in the learning process (Pintrich, 2000). Digital GBLEs offer a high degree of autonomy as students are free to establish their own in-game goals, make choices regarding how they proceed, and how they approach problems presented to them in the environment. To this end, learners can be supported in GBLEs in ways not possible in traditional settings given the ability to utilize stealth assessment (Shute, 2011) that can lead to customized, real-time scaffolds.

Thus, in order to advance the research momentum related to the academic benefits of GBLEs, it is important to begin to build sound GBLEs that provide the potential for learning and subsequent transfer of that learning outside the game environment. In addition, assessments that provide ecologically valid measures of transfer are also needed. These developments can then provide the context in order to study game scaffolds and mechanics that promote transfer and also characteristics of the players themselves that promote transfer. The current study is focused on the latter. The game employed in this study, *Crystal Island – Uncharted Discovery*, was developed with mechanisms, further described below, that provided the potential for learners to both learn and subsequently transfer their learning. The purpose of this study is to first describe a GBLE that was employed over four weeks in a fifth-grade classroom and was equipped with opportunities, if utilized, that would encourage learning and the transfer of science content knowledge. Second, the study reviews the process of integrating the GBLE into classrooms and then establishing an ecologically valid classroom measure of transfer. The primary research questions then focused on the extent to which cognitive and motivational factors, typically found within prevalent models of SRL, predicted successful transfer in a classroom activity external to the game environment. Given well-documented gender differences related to digital games (Borgonovi, 2016; Hamlen, 2010; Homer, Hayward, Frye, & Plass, 2012; Hsieh, Lin, & Hou, 2015), predictive models were also run separately by gender in order to develop an understanding of potential differences that may impact modifications for future development of GBLEs.

1.1. Theoretical background

Transfer has been defined as the productive use of prior learning in a novel context (Pan & Rickard, 2018). Bransford, Brown, and Cocking (2000) emphasize that transfer is an active rather than passive process and occurs more readily when knowledge is not overly contextualized. Unfortunately, often times learners do not see the applicability of skills learned in academic contexts with new contexts in which those skills might be applied. This problem likely arises from the fact that students perceive the skills they are learning as being too domain-specific or irrelevant in other contexts. Achieving transfer has long been one of the grand challenges of traditional education (Thorndike & Woodworth, 1901), but now it is at the forefront for digital learning environments as well. GBLEs, in particular, face scrutiny for their ability to promote transfer given skepticism about their legitimacy as learning environments and being perceived instead as motivational "fluff" (O'Neil, Wainess, & Baker, 2005). One recent review provided the answer of "maybe" when referring to whether skills and knowledge from digital games transfer to educational settings (Hacker, 2017).

Adams, Mayer, MacNamara, Koenig, and Wainess (2012) questioned the efficacy of narrative GBLEs to promote transfer when compared to more traditional approaches such as slideshow presentations. They suggested the need for extended gameplay for future investigations. Much of Mayer and colleagues' recent work has focused on how to encourage transfer using external aids outside of the gaming platform. Fiorella and Mayer (2012) found advantages for paper-based aids that prompted focus on important game features and underlying principles to promote transfer for college students on a basic circuit game. O'Neil, Chung, Kerr, Vendlinski, Buschang, and Mayer (2014) found that self-explanation prompts could be beneficial for transfer with middle school students by connecting game terminology with the curricular terminology (in this case mathematical concepts). Pilegard and Mayer (2016) utilized worksheets external to the game of *Cache-17*, a game designed to teach concepts related to electric circuits, and found those used in the middle of the game promoted transfer over a control condition and a pre-game worksheet condition. Moreno and Mayer (2005) found that explanatory feedback but not reflection increased transfer for college students who played a game focused on plant survival under different weather conditions. Similarly, Barzilai and Blau (2014) found advantages for an external scaffold with children aged 6–14 on problem-solving measures outside of a business simulation game. These studies are encouraging but more research is needed to examine transfer with self-contained games that do not require development of supporting materials or prompts.

There is limited evidence of transfer occurring with GBLEs in stand-alone contexts. Parong, Mayer, Fiorella, MacNamara, Homer, & Plass (2017) were successful in finding transfer from a custom-designed video game, *Alien Game*, and the executive function skill of shifting when players engaged in extended practice and practiced at a high level of challenge. Transfer was significantly correlated with performance in the *Crystal Island – Outbreak* game and students who completed the game also scored significantly higher on transfer measures (Nietfeld, Shores, & Hoffmann, 2014). Some GBLEs have the potential to promote transfer given that the environment offers problem-solving through an applied and exploratory approach followed by expert feedback (Schwartz, Chase, Oppezzo, & Chin, 2011). However, the challenge remains for students to gain a full understanding of the underlying structure of the targeted problems and content in order to be able to apply their newfound knowledge in a context outside of the game (Gick & Holyoak, 1983).

The extent to which learners learn academic content from a GBLE or make attempts to transfer knowledge may largely be dependent on exhibiting effective SRL. Prevalent cognitive and social-cognitive models of SRL (Pintrich, 2000; Winne & Hadwin, 1998; Zimmerman, 2000) emphasize cognitive, metacognitive, and motivational processes that impact learning within "episodes" experienced by the learner (Winne, 2010). Common across theories is the understanding that within these episodes learners engage in initial planning or forethought processes, then monitoring and regulation during task involvement, and finally some form of evaluation. SRL is malleable and should be conceptualized as a matter of degree where all learners exhibit some degree of self-regulation, however

more effective students have much greater skills. Moreover, SRL has been shown to be related to effective learning and achievement outcomes (Cleary & Kitsantas, 2017; Pintrich & De Groot, 1990; Yang, Chen, & Chen, 2018). From a SRL perspective, Hacker (2017) has argued the need for the games to assist the player in developing agency and to experience interactivity, choice, and control while at the same time providing guidance for the development of metacognitive skills that will facilitate transfer.

The emergence of GBLEs came with parallel assumptions that GBLEs' primary advantage over more traditional forms of instruction was their motivational appeal. However, the extent to which this is a valid empirical claim is yet to be fully determined as serious games have been shown to have stronger benefits for learning than motivation (Wouters, van Nimwegen, Oostendorp, & van der Spek, 2013). A number of motivation variables included within models of SRL have been increasingly introduced in the study of serious games. For instance, Mayer (2019) suggests that five classes of motivational theories that contribute to the motivational power of games: interest and value theories, self-efficacy and attribution theories, goal orientation theory, self-determination and intrinsic motivation theories, and social cue and embodiment theories. Variables from the first three of these theories were included in the current study. The first, situational interest, is considered to be a gateway to more long-lasting personal interest in that it can be the initiating effect that leads to continued participation and increased levels of interest over time (Schraw & Lehman, 2001). In GBLEs, situational interest has been shown to be higher for students who play video games versus engage in traditional instruction (Hwang, Wu, & Chen, 2012) and also higher within competitive and collaborative as opposed to individual conditions (Plass et al., 2013).

The second, self-efficacy, can be defined as one's confidence in their ability to plan and implement a course of action to solve a problem or accomplish a task (Bandura, 1997). Self-efficacy has been shown to be predictive of performance in GBLEs (Nietfeld et al., 2014) and improve as the result of playing GBLEs (Meluso, Zheng, Spires, & Lester, 2012; Sung & Hwang, 2013). The current study examined both a measure of self-efficacy for playing video games and also for learning science.

The third motivational variable is goal orientation that focuses on two primary orientations corresponding to mastery and performance (Ames & Archer, 1988; Dweck, 1986; Elliot & McGregor, 2001). Mastery-approach individuals are focused on gaining knowledge, increasing ability, and incrementally gaining skill. Performance-approach individuals are interested in performing well relative to their peers. In traditional learning environments unique benefits have been show by both orientations (Harackiewicz, Barron, & Elliot, 1998) as mastery goals have been associated with higher levels of cognitive strategy use (Wolters, Yu, & Pintrich, 1996), metacognitive strategy use (Wolters, 2004), interest (Wolters et al., 1996), and cognitive engagement and effort (Meece, Blumenfeld, & Hoyle, 1988). Performance goals, on the other hand, have been linked to various outcome measures of achievement across studies (Senko, Hulleman, & Harackiewicz, 2011). The role of goal orientation in GBLEs is not yet clearly understood, however one might posit that it could determine the emphasis one places on achieving points and in-game rewards versus gaining content mastery while also influencing the extent to which one persists in attempting to solve a complex problem.

In the current study, measures of prior content knowledge and problem-solving processes were also included to represent cognition. The activation of prior knowledge is critical to SRL (Alexander, Dinsmore, Parkinson, & Winters, 2011; Alexander, Schallert, & Hare, 1991) and the framing of that knowledge in the learning process itself (Engle, Lam, Meyer, & Nix, 2012) and the ability of the learner to abstract general principals and apply them to new settings (Gick & Holyoak, 1983) is critical for transfer of learning. Prior knowledge is critical to assess when examining SRL to account for learner differences in knowledge and also because prior knowledge may influence SRL and performance in unique ways across domains (Alexander et al., 2011). SRL and prior knowledge are intertwined in that the development of new knowledge is dependent upon effective self-regulation and the quality of self-regulation and performance is highly dependent upon prior knowledge. Problem-solving scenarios are central to GBLEs and require the learner to engage in metacognitive monitoring and regulation activities (Kim, Park, & Baek, 2009). Presentation of a general problem-solving model to work from has been shown to increase academic problem-solving performance for novices (Youssef-Shalala, Ayres, Schubert, & Sweller, 2014).

One area of investigation related to transfer in GBLEs with very little research up to this point is the impact of gender on the transfer of learning. What is known is that boys exhibit higher self-efficacy for games (Hartmann & Klimmt, 2011) and spend more time playing games than girls (Hamlen, 2010). Moreover, both boys and girls appear to benefit from GBLEs on content knowledge (Nietfeld et al., 2014). Yet, an open question for the field is how gender impacts the process of attempting to transfer learning outside of GBLEs. Do varied gender-related experiences with GBLEs impact the process of achieving transfer outside GBLEs?

Fig. 1. *Crystal Island – Uncharted Discovery.*

1.2. Current study

The current study describes efforts to measure and predict transfer from one GBLE that was integrated into classroom instruction. *Crystal Island – Uncharted Discovery* (see Fig. 1) is an immersive GBLE for 5th graders described more fully in prior work (Lester et al., 2014). In the game, students play the role of a shipwrecked passenger who is stranded on a volcanic island. The overall goal is to solve a series of quests (7 total) in order to find a means to get off of the island. The quests are self-contained game-like activities focusing on landform identification, map navigation, and modeling (topics aligned with state science curriculum), respectively, and are leveled based on difficulty. The students are free to complete the quests in any order they please; however, students must successfully complete the first level of all quests before engaging in any of the second level quests. A number of scaffolds are available to assist in problem solving including maps, landform experts who happened to be among the ship-wrecked crew, a tablet equipped with note-taking tools, a camera, a log to monitor quest completion and progress, a glossary of key landform and map skill terminology, and a problem-solving application.

Crystal Island – Uncharted Discovery is fully self-contained therefore external materials or cues to encourage transfer were not included in the study. The interface of the game encouraged the integration and application of conceptual knowledge within the quests and provided tools. For example, metacognitive prompts were provided using a simulated tablet to cue the students to check their understanding, keep notes, access additional information, communicate with scientists in the game, and keep track of tasks. The game employed in the current study was designed and tested by an inter-disciplinary team over the course of four years. The current study utilized an extended gameplay approach to report on the culminating integration of the game over a four-week time period.

Two primary research questions were explored to examine learner variables that predicted success on the transfer activity. *What motivational and cognitive variables predict transfer in a science GBLE?* Given the Wouters et al. (2013) meta-analytic finding that GBLEs have thus far failed to support motivation more than conventional instructional condition it is important to turn the directionality of influence around and examine how motivation impacts learning in GBLEs. Specifically, this study examined the unique contributions of motivational and cognitive variables, prevalent in models of SRL, in predicting transfer of learning outside the GBLE. It was not possible to measure all motivational variables important for SRL in the current study, however those chosen are generally considered to be fundamental within current models of SRL and critical when assessing motivation in GBLEs (Mayer, 2019; Pintrich, 2000; Winne & Hadwin, 1998; Zimmerman, 2000). In order to test this question a multiple regression approach was undertaken in order to examine the unique effects of prior knowledge, changes in knowledge that took place during the duration of gameplay, pretest knowledge of problem solving, and motivational variables including self-efficacy, goal orientation, and interest with a large sample of students. In

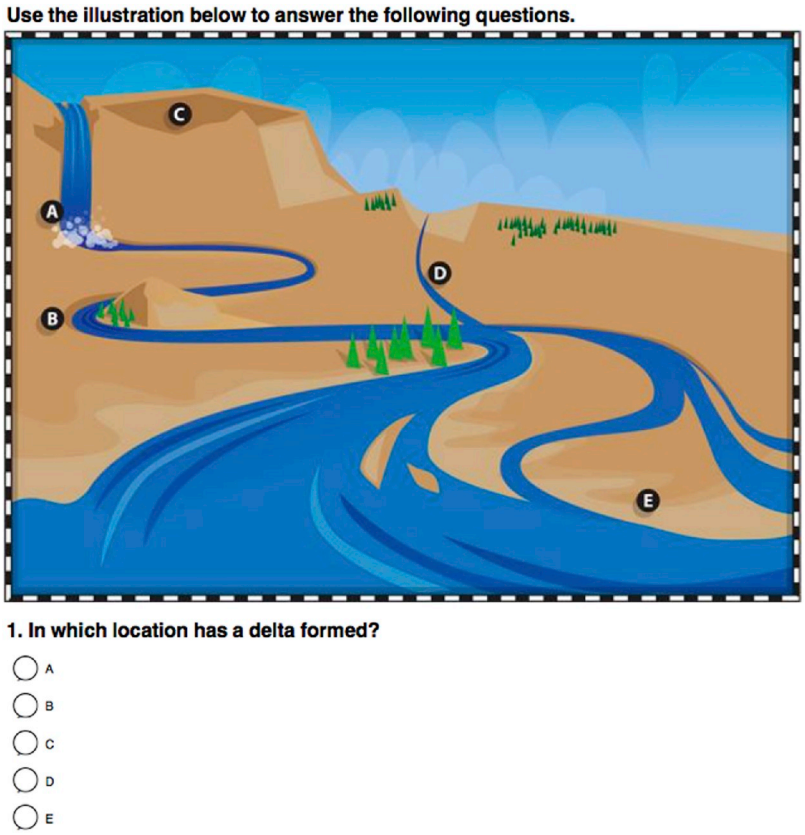

Fig. 2. Sample content knowledge item.

addition, the large sample and multiple regression approach allows for estimates of the amount of variation accounted for in the predictor variables as a whole to indicate how well transfer can be explained from multiple sources. Prevalent models of SRL argue for the contributions of each of these targeted variables in the learning process (Pintrich, 2000; Winne & Hadwin, 1998; Zimmerman, 2000). Multiple models were run to predict science content knowledge, in-game performance, and transfer to provide comparisons with regard to the relative ability overall to predict each outcome measure.

The second research question was *What role does gender play upon transfer in a GBLE?* Gender has played a significant role in the use, engagement, and learning processes explored in the serious games' literature (Hamlen, 2010; Homer, Hayward, Frye, & Plass, 2012). It is possible that through these different experiences and expectations of GBLEs that variations in pathways of academic learning would also emerge. Running separate models to predict transfer by gender will provide insight about learning in GBLEs that is currently missing from the literature. The existing literature suggests numerous differences by gender for gaming experience and motivation so it is reasonable to expect variation in predictive models. However, existing literature regarding potential gender differences in transfer of learning in either traditional or digital environments is scant and prevent the inclusion of firm hypotheses. These findings would have implications for the customization of scaffolds and design in the development of GBLEs.

2. Method

2.1. Participants

A total of 594 students (97% were between 10 and 11 years old; 52% female) across eight schools participated. Intact classrooms agreed to participate and include Crystal Island – Uncharted Discovery as part of their regular classroom instruction. The schools, teachers, and students chosen for the study had agreed to participate more broadly in a multi-year project aimed at developing *Crystal Island – Uncharted Discovery*. However, the specific students in the current study had not participated in the game prior to this study. The students were represented by the following ethnic breakdown: American Indian/Alaskan native 1.3%; Asian 7.7%; Black/African American 12.8%; Hawaiian/Pacific Islander .3%; Hispanic/Latino 5.2%; White 65.5%; Multiracial 7.1%.

2.2. Materials

Content Knowledge Test. This was measured with a research team-constructed, 19-item multiple-choice test that was based on state curriculum and designed to measure domain-related material integrated within Crystal Island – Uncharted Discovery. The same items were used as a test of prior knowledge ($\alpha = 0.80$) and as a posttest measure ($\alpha = 0.78$). To avoid test/retest learning no feedback was provided on item correctness on the pretest. The items ranged in requirement from knowledge/definitional level to application-level items (see Fig. 2 below). An additional variable was constructed to measure change from the pretest to the posttest as an indication of in-game learning. Additional examples of both knowledge/definitional items and an application item are provided in Appendix A.

Gaming Self-Efficacy. Video game self-efficacy was measured using a four-item scale ($\alpha = 0.79$). For example, one item asked, "I am certain that I can master any video game I play." The *Gaming Survey* was given as a pretest.

AGQ (Achievement Goals Questionnaire): This is a 12-item scale that measures achievement goal orientation in the form of four factors: mastery approach, mastery avoidance, performance approach, and performance avoidance (Elliot & McGregor, 2001). For this study the mastery approach ($\alpha = 0.86$) and performance approach ($\alpha = 0.83$) facets were included in the analysis.

Situational Interest. At the end of Crystal Island – Uncharted Discovery participants were asked to rate their interest in the game with 10 items adapted from the *Perceived Interest Questionnaire* (Schraw, 1997). For example, one item asks "Did you enjoy this game?"

Table 1
Procedures overview for classroom integration study.

Week	Planned Activities
Week 0	Pre-test in advance
Week 1	D1: Introduction, tutorial (introduce overall goal) D2: In-class landforms lesson with paper assessment D3: Game play landforms (enable landform quests)
Week 2	D1: In-class lesson maps and navigation D2: In-class lesson maps and navigation with paper assessment D3: Game play maps and navigation (enable map & navigation quests)
Week 3	D1: In-class lesson modeling D2: Game play modeling (enable modeling quests) D3: Game play modeling with paper assessment
Week 4	D1: Game play (enable multi-skill quest) D2: Lesson and work on Final Inquiry project D3: finish product and present D4: Post-test

Students responded on a 10 point Likert scale with one being "Not at all true of me" and 10 being "Very true of me." A composite score was created across items (α = 0.92).

Science Self-Efficacy. A science learning self-efficacy inventory was given as a pretest that included three items each for landforms (sample item: "I feel confident that I will be able to learn about landforms in science"; α = 0.70), map navigation (sample item: "I am sure that I can learn map navigation skills, such as using a compass, map grid, and map scale"; α = .64), and map models (sample item: "I am sure that I could use a model to represent a landform or a location"; α = .66). Each item was answered on a five-point Likert scale and a sum score (α = 0.82) was created from the three subscales.

Problem Solving. This was an eight-item test given as a pretest (α = 0.80) for identifying and applying problem-solving steps. The first four items asked students to correctly order steps of a general problem solving approach (Polya, 1957) and for the last four items students were provided with a fictitious problem (create a school map for a new student at your school) and asked to provide the correct order of approaches for solving the problem that represented steps of the general problem-solving model.

2.3. Procedures

The study took place over four weeks (see Table 1). Participating classroom teachers first attended a one-day professional development workshop to discuss plans for the integration of Crystal Island – Uncharted Discovery into their curriculum. The schedule included six 50-min classroom sessions playing the game that took place in parallel with six traditional classroom lessons that aligned with the curriculum in the game. While classroom effects could not be eliminated in the current study, steps were taken to minimize those effects. For example, gameplay within Crystal Island – Uncharted Discovery was self-contained and required no interaction with teachers. Moreover, for the supplemental in-class lessons, lesson plans and scripts were created and agreed upon by the participating teachers and project research team and made accessible to all teachers on a project website.

Creation of the transfer measure was undertaken by carefully matching the Crystal Island – Uncharted Discovery curriculum with the requirements of the task and presented as a Final Inquiry project completed during the last week of the study. The task required students to display mastery of landforms, map navigation, and map models. To ensure content validity participating classroom teachers were provided with the task beforehand and asked to provide feedback. The task directions asked students to draw an outline of an island on a large sheet of grid paper. They were provided with colored pencils and asked to include seven landforms on their island that included a mountain, river, waterfall, plateau, delta, lake, and tributary. Students were instructed that they could also include other landforms and landmarks as long as they included the seven required landforms. They were asked to consider the location of their landforms and also the accuracy of their drawings as they drew. For map navigation, students were asked to select three locations where they could hide secret treasures. However, they were instructed not to draw the treasures. Instead, they were asked to create three clues of increasing difficulty that they could give to their peers as a game. The instructions were as follows:

Clue #1: On the blank sheet of paper provided, write a clue or set of clues using at least one grid coordinate that could help a classmate find your first hidden treasure.

Clue #2: Using a landmark or landform as a starting point, write a clue or set of clues using compass directions and the map scale to help a classmate locate your second hidden treasure.

Clue #3: Using a landmark or landform as a starting point, write a clue or set of clues using compass directions, the map scale, and the map grid to help a classmate locate your third hidden treasure.

For map models, students were provided with a smaller grid and asked to draw a small model of their island. They were instructed

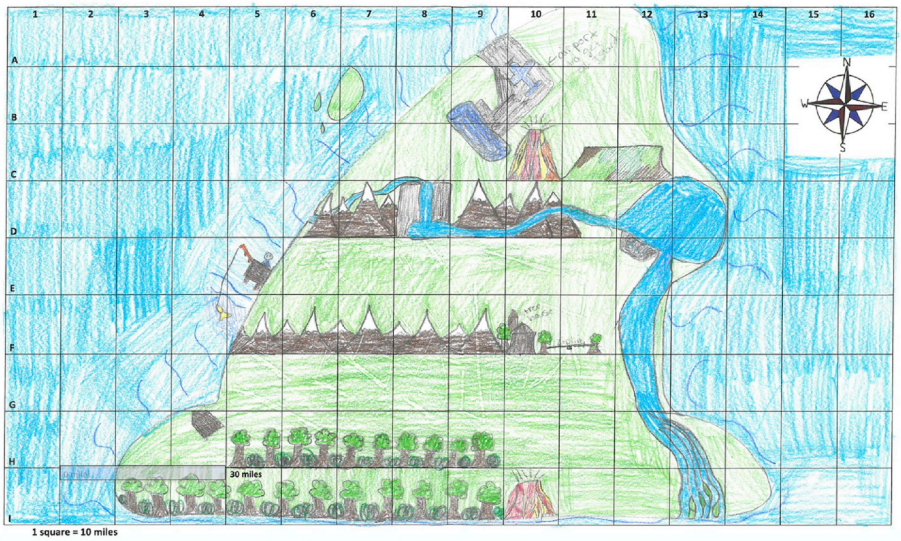

Fig. 3. Sample high-scoring map.

to recreate their island in smaller scale as accurately as possible, to include and label a symbol for each of their landforms, treasures, and any additional landmarks that were included, and to make sure that their hidden treasures were included on their small map. This smaller map functioned as an answer key to the map navigation game they would present to their peers. Students completed the entire activity during one class session.

After the maps were created students exchanged their large maps and clue sheets with a classmate while keeping their small model sheet hidden. Students then attempted to solve the clues provided to them by their peers and then ended the activity by exchanging the map model answer keys.

A graduate student oversaw a team of four teachers hired to score the maps. The group used a rigorous scoring and reliability process that included the creation of a scoring rubric. Each project was scored for the inclusion of landforms on the map and key, accurate drawing and location of landforms, and on the construction of accurate problems and answers for a total of 28 possible points ($M = 20.60$; $SD = 4.35$; range = 4–28). Two raters scored each map and key. Discrepancies between raters were discussed until arriving at an agreed upon score. These results suggest that overall students were able to transfer a substantial amount of the content from the game to an external task. Students were most successful in constructing components of their map (e.g., including accurate landforms and accurate key) as they scored 81% of the total possible points. Their greatest challenge came in creating accurate map navigation problems and clues for their peers to complete as they scored only 45% of the total possible points. A sample student island (Fig. 3) and key (Fig. 4) are provided below.

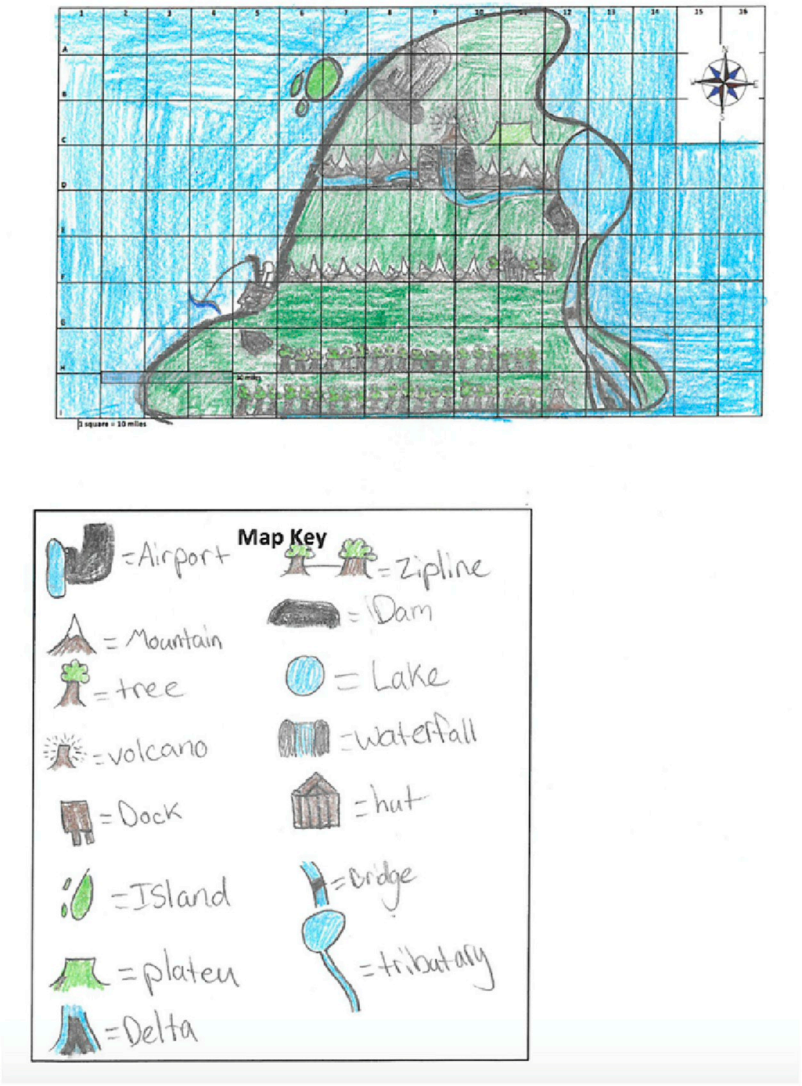

Fig. 4. Sample high-scoring key.

2.4. Data analysis

The primary data in the current study consisted of responses from self-report inventories, in-game trace measures of quests completed, performance scores on a science content test, and scores on a performance-based measure of transfer completed in a classroom context. For the measure of transfer, rubrics were created and subsequently applied using multiple raters to determine scores. Primary analysis was conducted using multiple regression in order to examine the unique effects of prior knowledge, changes in knowledge that took place during the duration of gameplay, pretest knowledge of problem solving, and motivational variables including self-efficacy, goal orientation, and interest with a large sample of students. The large sample and multiple regression approach allowed for estimates of the amount of variation accounted for in the predictor variables as a whole to indicate how well transfer can be explained from multiple sources. Multiple models were run to predict science content knowledge, in-game performance, and transfer to provide comparisons with regard to the relative ability overall to predict each outcome measure.

3. Results

Descriptive statistics are provided for the major study variables in Table 2.

An examination of the impact of gameplay on science content knowledge was first accomplished using pre and post science tests. Results revealed significant gains (Cohen's $d = 0.71$) in science content knowledge from pre ($M = 11.93$, $SD = 3.97$) to posttest ($M = 13.81$, $SD = 3.52$).

Both boys ($M_{\text{pretest}} = 12.32$; $SD = 3.86$; $M_{\text{posttest}} = 14.21$; $SD = 3.21$; $t = -12.49$; $p < .001$, Cohen's $d = 0.75$) and girls ($M_{\text{pretest}} = 11.57$; $SD = 4.04$; $M_{\text{posttest}} = 13.45$; $SD = 3.74$; $t = -11.74$; $p < .001$, Cohen's $d = 0.67$) showed significant increases in science content knowledge after gameplay. Boys scored significantly higher on the content posttest ($t = 2.65$; $p = .008$, Cohen's $d = 0.21$), however, no significant differences remained between genders on posttest knowledge when controlling for pretest scores. Yet, boys completed more game quests even after controlling for game time ($p < .001$; $\eta_p^2 = 0.03$). No significant differences were found between boys ($M = 20.10$; $SD = 4.25$) and girls ($M = 21.05$; $SD = 4.40$) on the transfer task.

3.1. What cognitive and motivational variables predict transfer in a science GBLE?

A multiple regression analysis was conducted to predict scores on the transfer task. The model included the following predictor variables: science prior knowledge, change in knowledge from the pre to posttest, problem solving, video game self-efficacy, self-efficacy for science content, interest for *Crystal Island – Uncharted Discovery*, mastery approach goal orientation, and performance approach goal orientation. A number of assumptions regarding the analysis and data were tested. First, data was examined for outliers using standardized residuals and no cases were found (Std. Residual Min = −3.17, Std. Residual Max = 1.87) using 3.29 and −3.29 as thresholds (Field, 2013). Next, tests for the assumption of collinearity indicated no concerns (Tolerance range from 0.46 to 0.97; VIF range from 1.05 to 2.20). Furthermore, the data met the assumption of non-zero variances (Video Game Self-Efficacy variance = 13.84, Problem-Solving variance = 1.40, Mastery Approach variance = 20.65, Performance Approach variance = 38.73, Science Prior Knowledge variance = 15.75, Knowledge Change variance = 7.23, Science Self-Efficacy variance = 15.62, Crystal Island interest variance = 63.33; Map Transfer task variance = 18.94).

When predicting *transfer* the model was significant $F_{(8,559)} = 11.13$, $R^2 = 0.137$, $p < .001$ (see Table 3). Within the model, science prior knowledge ($\beta = 0.363$, $p < .001$), change in knowledge ($\beta = 0.170$, $p < .001$), and pre-study video game self-efficacy ($\beta = -0.101$, $p = .011$) were all significant predictors. Both prior knowledge and change in knowledge were positive predictors of transfer whereas video game self-efficacy was a negative predictor. None of the motivational variables were significant predictors of transfer.

3.2. What role does gender play upon transfer in a GBLE?

Three multiple regression models were run for each gender to predict posttest science content knowledge, in-game performance as measured by completed quests, and performance on the 'create an island' transfer task separately by gender. The following prediction variables were included in each model: pretest content knowledge, science self-efficacy, mastery approach, performance approach,

Table 2
Means and standard deviations of study variables.

Variable	Boys	Girls	Total
Prior Knowledge	12.32 (3.86)	11.57 (4.04)	11.93 (3.97)
Posttest Content	14.21 (3.21)	13.45 (3.74)	13.81 (3.52)
Gaming Self-Efficacy	16.53 (3.37)	13.71 (3.52)	15.06 (3.72)
Mastery Approach	20.53 (4.60)	20.69 (4.50)	20.61 (4.54)
Performance Approach	13.96 (6.05)	11.93 (6.23)	12.90 (6.22)
Situational Interest	38.60 (8.54)	38.38 (7.40)	38.48 (7.96)
Science Self-Efficacy	28.94 (3.13)	28.60 (3.32)	28.76 (3.23)
Problem Solving	4.20 (2.48)	4.27 (2.55)	4.24 (2.52)
Game Quests Completed	5.24 (2.36)	4.27 (2.43)	4.73 (2.44)
Transfer Task	20.10 (4.25)	21.05 (4.40)	20.60 (4.35)

Table 3
Multiple regression predicting transfer scores.

	(N = 567)		
Predictors	B	SE	β
Science prior knowledge	.40**	.06	.36**
Change in knowledge	.27**	.08	.17**
Video game Self-Efficacy	-.12*	.05	-.10*
Problem Solving	.11	.08	.06
Science Self-Efficacy	-.08	.06	-.06
Performance Approach	.03	.03	.05
Mastery Approach	.01	.04	.02
Interest	.03	.02	.05
R^2		.14	

Note. * < 0.05; ** < 0.01.

interest, problem solving, and video game self-efficacy. Assumptions were examined for each model as in the overall regression above. Two outliers were removed in the model for boys predicting posttest science content knowledge. Two outliers were also removed in the model for girls predicting performance on the transfer task. The models reported below meet the assumptions for outliers, collinearity, and of non-zero variances.

The results (see Tables 4 and 5) revealed that each of the six models were significant. For girls: predicting posttest science content knowledge, $F_{(7,289)} = 62.79$, $R^2 = 0.603$, $p < .001$; predicting completed quests, $F_{(7,289)} = 14.03$, $R^2 = 0.254$, $p < .001$; predicting transfer, $F_{(7,287)} = 7.33$, $R^2 = 0.131$, $p < .001$ (see Table 4). For boys: predicting posttest science content knowledge, $F_{(7,261)} = 57.26$, $R^2 = 0.606$, $p < .001$; predicting completed quests, $F_{(7,263)} = 13.84$, $R^2 = 0.269$, $p < .001$; predicting transfer, $F_{(7,263)} = 5.66$, $R^2 = 0.131$, $p < .001$ (see Table 5). Science prior knowledge was a significant predictor in each of the six models. Interest was a consistent significant predictor across all three models for girls but was not a significant predictor in any of the models for boys. For both genders the amount of variance accounted for in the models systematically decreased moving from posttest content knowledge to in-game performance and finally to the transfer task. Also, of note, was that problem-solving was a significant predictor in all models with the exception of predicting transfer for boys. Neither self-efficacy variable was a significant predictor in the models with the exception that science self-efficacy was a significant *negative* predictor ($\beta = -0.09$; $p < .05$) of content knowledge for girls. Similarly, performance approach goal orientation was not a significant positive predictor of performance in any of the six models but was a significant *negative* predictor of quests completed for boys. Mastery approach, on the other hand, was a significant predictor of content knowledge for both boys and girls.

4. Discussion

This study focused on player characteristics within a SRL framework that predicted transfer of learning from a GBLE focused on science in 5th grade classrooms. Students who played *Crystal Island – Uncharted Discovery* over four weeks demonstrated significant learning gains. These gains coupled with the creation and implementation of a transfer measure that garnered support from the participating teachers was encouraging. The study contributes to the existing GBLE literature in that it is one of the first to examine how key motivational and cognitive SRL variables predict transfer with a large sample of students ($N = 594$) after an extended period of classroom integration. It is also one of the first studies to examine these predictors of transfer in GBLEs with a particular focus on gender. Major findings and implications for the major study questions related to predicting transfer follow below.

From a criterion-based perspective, students appeared to be successful in transferring their knowledge of landforms and application skills in creating a map and corresponding scale. Understandably, however, students had more difficulty creating the map navigation game for their peers. While not measured in the current study, one might posit that students who were "surface focusers" may have needed additional support either within or outside of the game or both in order to be successful in applying their knowledge in the

Table 4
Summary of Hierarchical Regression Analysis for Variables Predicting Performance and Transfer by girls (N = 296).

Variable	Model 1 Science Content Knowledge			Model 2 Quests Completed			Model 3 Transfer		
	B	*SE B*	*β*	*B*	*SE B*	*β*	*B*	*SE B*	*β*
Science Prior Knowledge	.51	.04	.56**	.21	.04	.34**	.20	.07	.19**
Problem Solving	.31	.06	.22**	.13	.06	.14*	.21	.11	.13*
Video game self-efficacy	-.01	.04	-.01	.02	.04	.02	-.07	.07	-.06
Science Self-Efficacy	-.08	.04	-.09*	.02	.04	.04	.03	.07	.03
Performance Approach	-.01	.02	-.01	-.03	.02	-.09	.06	.04	.10
Mastery Approach	.14	.04	.17**	.04	.03	.07	.07	.06	.08
Interest	.04	.02	.08*	.04	.02	.12*	.08	.03	.14*
R^2		.60			.25			.15	.

Note: $*p < .05$. $**p < .01$.

Table 5
Summary of Hierarchical Regression Analysis for Variables Predicting Performance and Transfer by boys (N = 270).

Variable	Model 1 Science Content Knowledge			Model 2 Quests Completed			Model 3 Transfer		
	B	*SE B*	*β*	*B*	*SE B*	*β*	*B*	*SE B*	*β*
Science Prior Knowledge	.56	.04	.69**	.22	.04	.35**	.38	.08	.35**
Problem Solving	.12	.06	.10*	.14	.06	.15*	.07	.12	.04
Video game self-efficacy	.05	.04	.05	.07	.04	.10	-.06	.07	-.05
Science Self-Efficacy	.01	.03	.01	.03	.03	.06	-.10	.06	-.09
Performance Approach	-.02	.02	-.03	-.05	.02	-.12*	.01	.04	.01
Mastery Approach	.06	.03	.09*	.04	.03	.07	-.02	.06	-.02
Interest	.02	.02	.05	.02	.02	.07	.00	.03	.00
R^2		.61			.27			.13	

Note: $*p < .05$. $**p < .01$.

higher-level task of creating the navigation game (Siler & Klahr, 2015). The extent to which transfer of learning was successful in the current study is likely due to the fact that *Crystal Island – Uncharted Discovery* provided simulated performance-based quests at two levels for each content area. The quests functioned to break the larger problem down into manageable units and provide frequent feedback (Shores, Hoffmann, Nietfeld, & Lester, 2012).

The first research question examined the ability of motivational and cognitive variables, prevalent in models of SRL, in their ability to predict transfer. Surprisingly, across the sample as a whole, no motivation variables were significant positive predictors of transfer. Relatedly, meta-analysis by Wouters et al. (2013) found that serious games were not more motivating than conventional instructional methods. However, the current study approached motivation from a different perspective: the impact of players' motivation on the transfer of learning – but with the same general finding that motivation variables, at least those included in this study, appeared to have little direct impact on the transfer of learning. The motivational variables employed in this study were certainly not exhaustive but represented three of the five dimensions of motivation that contribute toward the motivational power of games according to Mayer (2019). Following prevalent approaches by prior studies, all of the motivational measures were self-report. Given the known limitations of such measures (Winne, 2010) trace-based measures and physiological indicators of motivation may have yielded different results and should be considered in future studies. It is also possible that motivational variables such as self-efficacy, goal-orientation, and interest are critical to engage learners in GBLE activities and may influence transfer in more indirect pathways, but that the development of knowledge structures allowing for cognitive flexibility is primary in the direct relationship of knowledge transfer (Spiro, Vispoel, Schmitz, Samarapungavan, & Boerger, 1987).

The inverse relationship between video game self-efficacy and transfer could be interpreted in multiple ways. One explanation could be a general level of overconfidence by those students who have had more experience playing video games. Another potential explanation could be a disconnect in expectations for commercial versus serious games. Performance in *Crystal Island – Uncharted Discovery* emphasized the learning, understanding, and application of content knowledge over and above action-based motor skills developed by the players. Even though the game was being presented in a school setting, this was likely not well understood by students when they considered their reference points when making self-efficacy judgments. The finding of change in knowledge as a significant predictor suggests that students who made the most learning gains within the game were also those most likely to transfer their knowledge outside of the game. This finding is promising for the potential of games such as *Crystal Island – Uncharted Discovery* to facilitate transfer. Numerous other unaccounted for effects, namely the parallel classroom instruction, could have impacted this effect, yet this study presents initial findings that should lead to more controlled follow up studies. Indeed, however, only 14% of the total variance for predicting transfer performance was accounted for in the model.

The second research question in the current study examined the separate profiles of transfer-related performance by gender. Girls performed as well as boys on the transfer task even though they had completed significantly less game quests and had significantly lower posttest content scores. Regression models for both genders were systematically less able to predict performance moving from content knowledge, to game performance, to transfer indicating the challenge facing researchers in identifying successful pathways to transfer. Not surprisingly, prior knowledge was a consistently strong predictor of content knowledge, in-game performance, and transfer across genders. The finding that stands out the most is the fact that even through there were no differences between genders for interest ($t = 0.32$, $p = .75$), interest was a consistent predictor of performance for girls across all three regression models but not for boys. In prior studies, positive relationships between science interest and performance have been found for *both* genders and cultures, although smaller in magnitude than mathematics (Evans, Schweingruber, & Stevenson, 2002). Separate analyses by gender in the Author (20xxb) study failed to show situational interest as a significant predictor of in-game performance in either model. Given the limited prior work in this area, it appears that direct comparisons to other studies predicting transfer by gender, particularly in GBLEs, is unavailable to date. If findings from the current study are replicated and generalize across contexts the implications imply a need to assess interest in the GBLE early on in the user experience, particularly for girls. This also includes the need to carefully consider features of the GBLE itself during the process of development that would likely promote interest for girls.

Results of the current study reveal a challenge for predicting transfer for boys as only prior knowledge was a significant in the model. Thus, researchers must delve deeper and consider more trace-based measures or other real-time data-tracking methods that are more sensitive to the process of transfer during the duration of time students are playing the game.

Interestingly, problem-solving was a significant predictor of content knowledge and quests completed for both genders yet was not

significant for either gender in predicting transfer. The 'Create an Island' transfer task was essentially a problem-based learning activity that required effective implementation of steps from the general problem-solving model assessed by the problem-solving pretest. Nevertheless, the problem-solving measure appeared to have a greater relative impact on content knowledge gains and in-game performance. The negative relationship between girls' science self-efficacy and their content knowledge gain is counter to limited prior research from similar GBLE contexts (Bergey, Ketelhut, Liang, Natarajan, & Karakus, 2015). Girls' judgments may have been in reference to more typical learning environments rather than from a GBLE in which they have less prior experience than their male peers. Similarly, performance approach was not a significantly positive predictor of performance in any of the six models but was a significantly *negative* predictor of quests completed for boys. One possibility for this finding is the lack of norm-based performance feedback. *Crystal Island – Uncharted Discovery* did include mastery-based recognition in the form of levels of trophies, however these did not provide performance-approach feedback allowing for comparison of performance relative to others. An interesting potential follow up might consider adding real-time measures that provide comparative indices between players that promote competition (Plass et al., 2013). Mastery approach, on the other hand, was a significant predictor of content knowledge for both boys and girls.

4.1. Implications for educators and designers of educational games

In addition to those discussed above, there are a number of implications from this study for educators and designers of educational games to consider. With regard to the negative relationship between video-game self-efficacy and transfer, educators may consider measures that ask students to consider their efficacy to *learn and apply* academic content outside of the game environment. If the negative relationship persists with these, presumably, more sensitive measures, one approach would be to develop mechanisms in the game environment to periodically check for calibration, or the ability to match one's confidence with performance, and to subsequently deliver appropriate scaffolds to increase calibration. Another important implication relates to the strong influence revealed by both prior knowledge and the change in knowledge during gameplay to predict transfer. Given this finding, developers of education games would be encouraged to work closely with classroom teachers to consider means to develop knowledge that can be 'flexibly packaged' within the game narrative with the intent of increasing the transfer of that knowledge outside the game. Relatedly, findings in the current study would support game designers including elements to encourage mastery approaches to learning in the game. Finally, given findings such as interest predicting girls' performance across the board, developers should be encouraged to work closely with teachers to ensure intake of diagnostic information early on in game interaction in order to make needed adaptations. This approach would likely also get at meaningful patterns that emerge beyond that of gender that were not accounted for in the current study.

4.2. Limitations

While the current study makes a unique contribution to the literature with a rigorously designed transfer measure integrated in an actual classroom, the major limitation was the lack of a comparison condition that did not employ a GBLE. Use of a comparison condition would have allowed for a more direct measure of the impact of a GBLE on transfer relative to 'business as usual' science instruction. Follow on studies should consider experimental designs following best practices for assessing GBLEs (All, Castellar, & Van Looy, 2016) by creating and describing comparable experimental and control conditions that vary only in the use of the GBLE itself.

4.3. Future studies

Future studies should examine more closely the variables that directly impact transfer in GBLEs. This will require a greater degree of control across conditions to isolate targeted components that may influence transfer of learning. Given the critical importance for students to be able to transfer their learning widely across contexts, future studies should also test the generalization of the current findings across game platforms and content domains and also to examine the mechanisms for transfer more deeply. The value and sustainability of serious game depends largely on the efficacy of such games to facilitate the display of student learning outside of the gaming context.

Author contributions

John L. Nietfeld: Conceptualization; Data curation; Formal analysis; Funding acquisition; Investigation; Methodology; Project administration; Resources; Software; Supervision; Validation; Visualization; Roles/Writing - original draft; Writing - review & editing.

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Appendix A

Examples of knowledge/definitional items:

11. Which of the following is NOT a model?

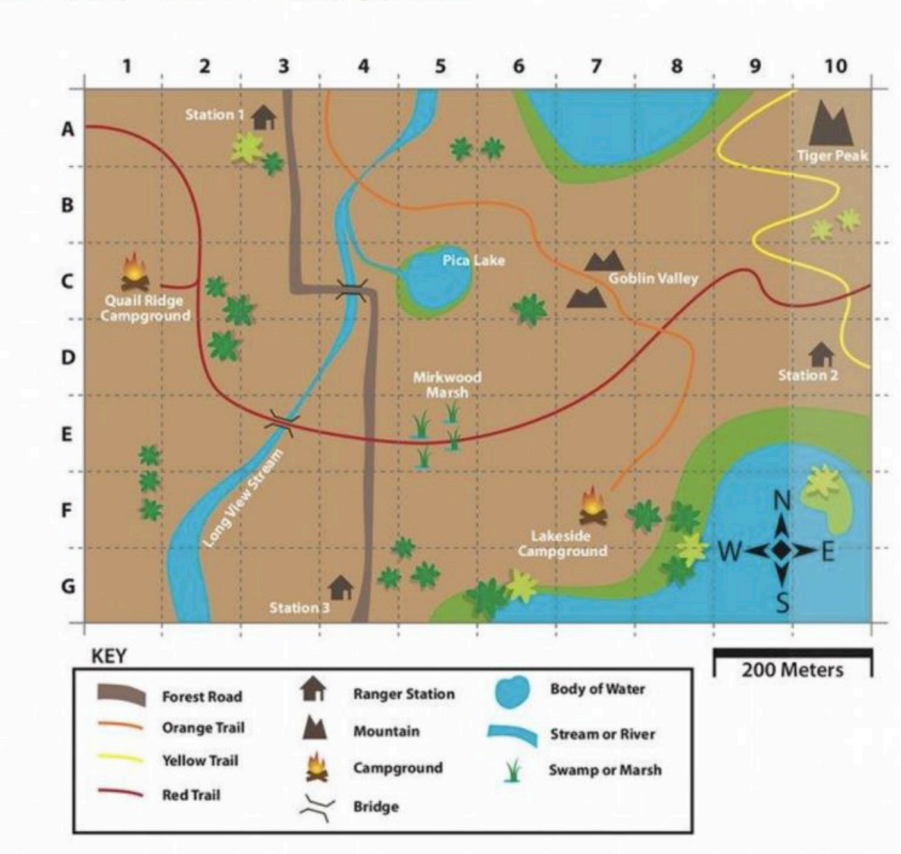
- a globe
- a doll
- a waterfall
- a map

12. Which of the following is a person who makes maps?

- Geographer
- Surveyor
- Cartographer
- Volcanologist

Example of an application item:

Use this map to answer the following questions.



16. From Ranger Station 3, travel north for 400 meters and then east for 100 meters. Where are you?

- Mirkwood Marsh
- Goblin Valley
- Pica Lake
- Ranger Station 2

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